Mould guide

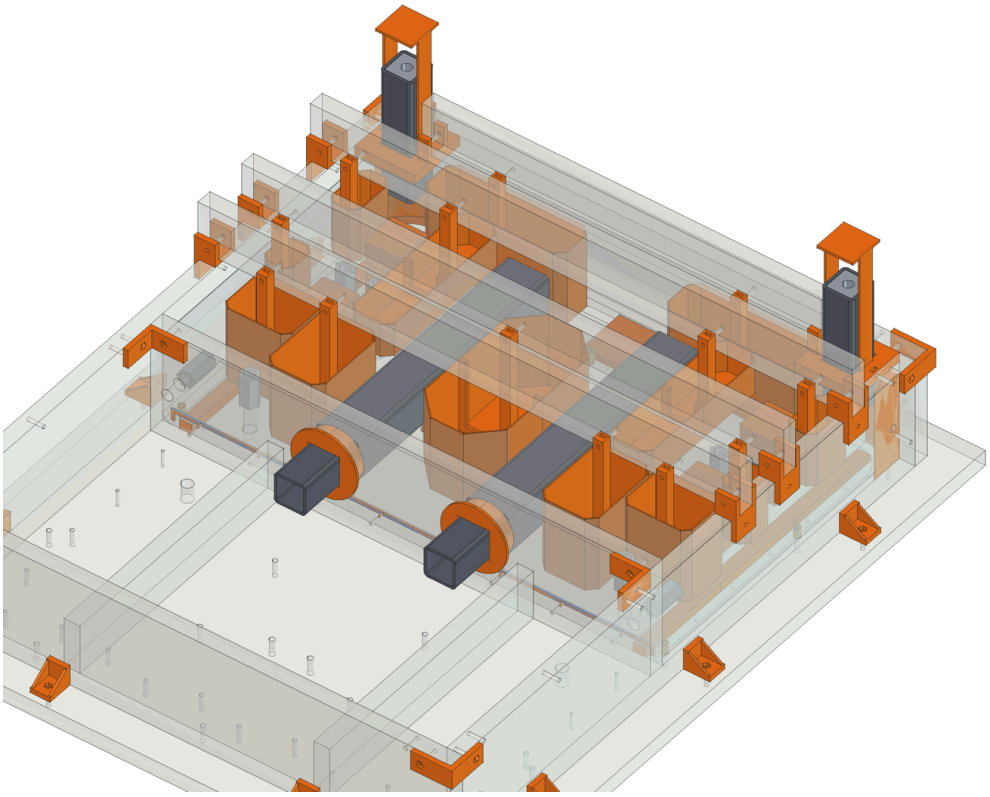
DIY CNC epoxy-granite mill · October 2026

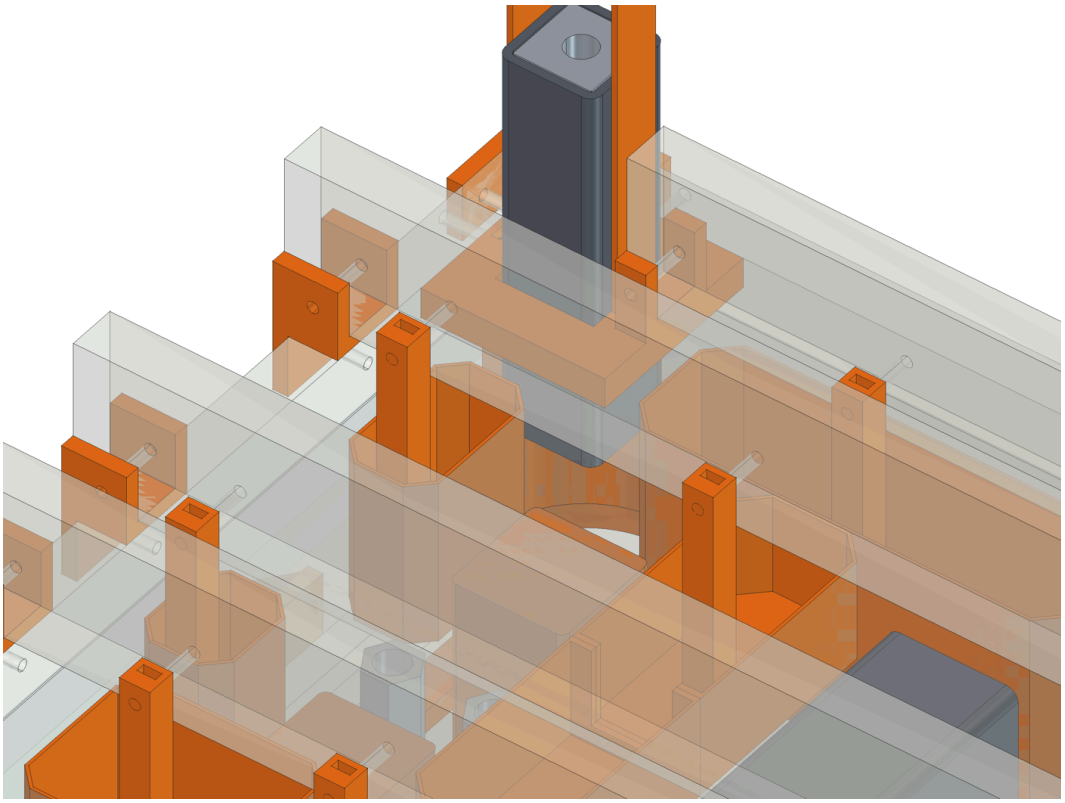
The moulds are melamine boards that bolt together with printed brackets, and every printed piece bolts or pins to its board from outside. Nothing is glued and nothing goes through a cast face, so the moulds come apart cleanly and go back together the same way for the next casting.

How it works

- **Walls stand 25 mm above the pour line.** The brackets, saddles and leg holders bolt through that band, so no bolt passes through a face that gets cast against. Fill to the line, not to the top.
- **Corner brackets** hold each outside corner square in that band; **floor brackets** hold each wall foot from outside and bolt down through the floor. Each bracket has its nut pressed into a hex pocket, and its holes are a drill guide: clamp it in place and drill through it.
- **Printed pieces bolt on from outside.** An M4 bolt (M3 on thin pieces) goes through the board into a nut built into the piece. The nut drops into a pocket in the face against the board and slides over under the hole, so nothing opens onto a cast face. Fit the nuts before the pieces go on.
- **Thin pieces have pins.** The gasket-groove beads and the shallow end of the gutters have 3 mm pins that push into 1/8 in holes. The casting presses them onto the board.
- **Pieces cut to fit the printer** have a peg and socket on the cut, and their own fixings.
- **Bridges** stand on edge in saddles that hook over the wall tops; each core has a post that bolts to the side of a bridge, which sets its height and stops it floating.
- **Fits** are 0.15 mm a side: snug, but they go together by hand. Waxed pieces come out after curing.

Measure your board first. The saddles, cradles and drawings are for 19.05 mm board (Home Depot's 3/4 in white melamine is a true 0.750 in; other "3/4 in" melamine can be 18 mm). If yours is different, set `mould_board` in `model/build_cnc.py` and re-run it: the STLs, hole maps and this guide all update.





What to buy

- **Melamine board**, 19 mm (3/4 in): about 2.6 m2 of boards in all, so 2 sheets of 4 x 8 ft with room for mistakes. Have the store rip it into strips at the widths in the cut list.
- **Hardware** (zinc-plated is fine; grease anything that touches the mix):

Item	Qty
M3 nut	32
M3 washer	44
M3 x 10 bolt	12
M3 x 25 bolt	20
M4 nut	166
M4 washer	166
M4 x 30 bolt	134
M4 x 35 bolt	28
M4 x 70 bolt	4
M5 x 20 screw, greased (holds an insert)	29
M6 x 20 screw, greased (holds a pad)	6
M6 x 20 screw, greased (holds an insert)	6
1/2 in plain steel rod, greased (forms a beam bolt hole), 198 mm	8
1/2-13 x 1-1/2 in bolt + nut, greased (forms the dowel hole)	2
1/2-13 x 1-1/2 in bolt, greased (holds a draw-rod coupler)	2
1/2-13 x 2-1/2 in bolt, greased (holds a coupler)	8
3/8-16 x 2-1/2 in bolt, greased (holds the top plate)	4
5/8 in plain steel rod, greased (forms a tie-rod hole), cut to 140 mm, 243 mm, 377 mm, 377 mm	2
Block of 2x4, 52 mm long: four under each upright mould (the posts stand on the bench)	8

Buy about 10 % spare of the small bolts, nuts and washers.

Cut list

Sizes are length x height (or width) of the face, in mm. Boards marked "profile" are cut to the outline in their drawing.

Mould	Board	Qty	Size (mm)	Notes
Base	Base floor	1	718 x 843	both halves are cast on it
Base	Base front wall	1	638 x 135	
Base	Base rear wall	1	638 x 135	
Base	Base side wall, left	1	724 x 135	
Base	Base side wall, right	1	724 x 135	
Base	Base partition	1	600 x 135	rear half only; comes out before the front half
Base	Base partition strut	2	326 x 60	offcut, stands on edge; holds the partition foot (rear half only)
Base	Base bridge y=-190	1	651 x 60	
Base	Base bridge y=-58	1	651 x 60	
Base	Base bridge y=100	1	651 x 60	

Mould	Board	Qty	Size (mm)	Notes
Base	Base bridge y=170	1	651 x 60	
Base	Base bridge y=280	1	651 x 60	
Base	Base leg bridge y=-310	1	480 x 60	
Base	Base leg bridge y=344	1	480 x 60	
Uprights	Upright floor	2	198 x 575	
Uprights	Upright side plate, outer	2	495 x 271 (profile)	cut to the profile
Uprights	Upright side plate, inner	2	495 x 271 (profile)	cut to the profile
Uprights	Upright back board	2	80 x 271	
Uprights	Upright front board	2	80 x 312	ends cut square
Uprights	Upright bridge	2	131 x 108	
Beam	Beam floor	1	648 x 258	
Beam	Beam top wall	1	568 x 165	
Beam	Beam bottom wall	1	568 x 165	
Beam	Beam end wall, left	1	140 x 165	
Beam	Beam end wall, right	1	140 x 165	

Drilling

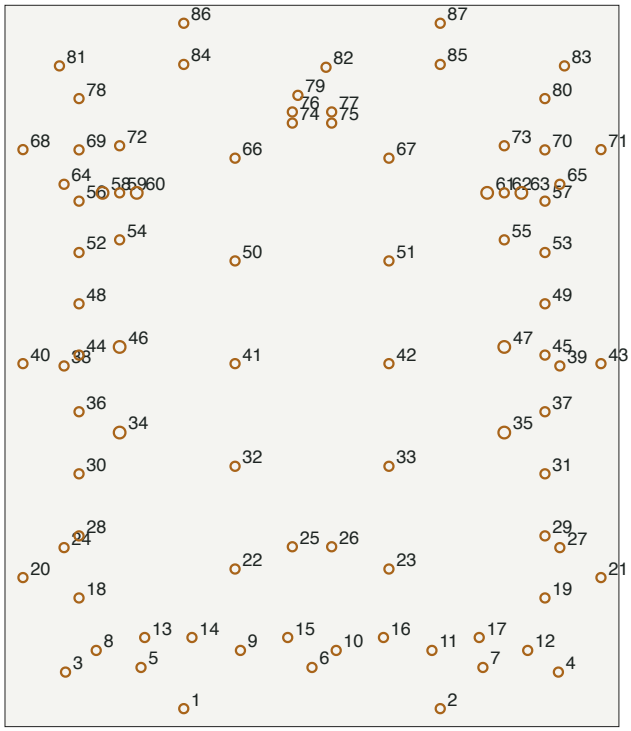
- Drill every hole from the **cast side** (the face the mix touches), so any tear-out is on the outside. The drawings and tables are seen from that side: x from the left edge, y from the bottom edge. For a wall the bottom edge sits on the floor; for a floor, the bottom of the drawing is toward the front wall (or the beam's bottom wall). Left and right are as you look at the cast face, so a floor seen from above is mirrored compared with the machine.
- Bridges are drawn from the top (outside).
- Use the printed brackets as drill guides where they sit; the tables give every hole anyway.

Hole	Drill	For
3.2 mm	1/8 in	foot-gasket-groove, motor-relief, split-gasket-groove
3.4 mm	9/64 in	corner-L, corner-R, gutter-L, gutter-R
4.5 mm	11/64 in	back-edge-1, core post, corner bracket, corner-L, corner-R, floor bracket, gutter-L, gutter-R, leg holder, saddle, slope-pocket--60, slope-pocket-40, socket plug, toe-filler, trough, tube locator
5.5 mm	7/32 in	M5 insert screw
6.5 mm	17/64 in	M6 insert screw
9.6 mm	25/64 in	3/8 dowel
10 mm	25/64 in	3/8-16 bolt into the top plate

Hole	Drill	For
13.1 mm	33/64 in	1/2 core rod, 1/2-13 bolt, base dowel hole
13.5 mm	17/32 in	1/2-13 bolt into coupler
16.3 mm	41/64 in	5/8 core rod
63.5 mm	2-1/2 in hole saw	hole saw

Base: hole maps

Base floor



0,0 · 718 x 843 mm · seen from the cast side

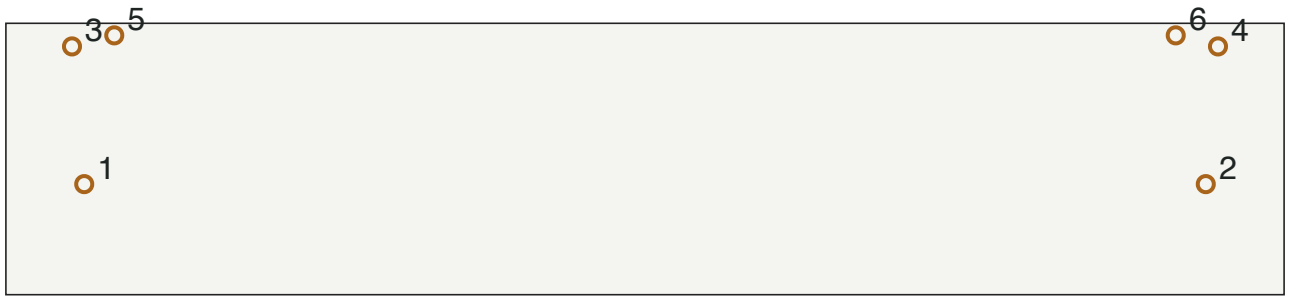
#	x	y	Hole	For
1	209	21	4.5	floor bracket
2	509	21	4.5	floor bracket
3	71	64	3.4	corner-R
4	647	64	3.4	corner-L
5	159	69	5.5	M5 insert screw
6	359	69	5.5	M5 insert screw
7	559	69	5.5	M5 insert screw

#	x	y	Hole	For
8	107	89	4.5	trough
9	275	89	4.5	trough
10	387	89	4.5	trough
11	499	89	4.5	trough
12	611	89	4.5	trough
13	163	104	4.5	trough
14	219	104	4.5	trough
15	331	104	4.5	trough
16	443	104	4.5	trough
17	555	104	4.5	trough
18	87	150	4.5	gutter-R
19	632	150	4.5	gutter-L
20	21	174	4.5	floor bracket
21	697	174	4.5	floor bracket
22	269	184	5.5	M5 insert screw
23	449	184	5.5	M5 insert screw
24	69	209	5.5	M5 insert screw
25	336	210	6.5	M6 insert screw
26	382	210	6.5	M6 insert screw
27	649	209	5.5	M5 insert screw
28	87	223	4.5	gutter-R
29	632	223	4.5	gutter-L
30	87	295	3.4	gutter-R
31	632	295	3.4	gutter-L
32	269	304	5.5	M5 insert screw
33	449	304	5.5	M5 insert screw
34	134	344	13.5	1/2-13 bolt into coupler
35	584	344	13.5	1/2-13 bolt into coupler
36	87	368	3.4	gutter-R
37	632	368	3.4	gutter-L
38	69	421	5.5	M5 insert screw
39	649	421	5.5	M5 insert screw
40	21	424	4.5	floor bracket
41	269	424	5.5	M5 insert screw
42	449	424	5.5	M5 insert screw

#	x	y	Hole	For
43	697	424	4.5	floor bracket
44	87	434	3.4	gutter-R
45	632	434	3.4	gutter-L
46	134	444	13.5	1/2-13 bolt into coupler
47	584	444	13.5	1/2-13 bolt into coupler
48	87	494	3.4	gutter-R
49	632	494	3.4	gutter-L
50	269	544	5.5	M5 insert screw
51	449	544	5.5	M5 insert screw
52	87	554	3.4	gutter-R
53	632	554	3.4	gutter-L
54	134	569	4.5	socket plug
55	584	569	4.5	socket plug
56	87	614	3.4	gutter-R
57	632	614	3.4	gutter-L
58	114	624	13.5	1/2-13 bolt into coupler
59	134	624	9.6	3/8 dowel
60	154	624	13.5	1/2-13 bolt into coupler
61	564	624	13.5	1/2-13 bolt into coupler
62	584	624	9.6	3/8 dowel
63	604	624	13.5	1/2-13 bolt into coupler
64	69	634	5.5	M5 insert screw
65	649	634	5.5	M5 insert screw
66	269	664	5.5	M5 insert screw
67	449	664	5.5	M5 insert screw
68	21	674	4.5	floor bracket
69	87	674	3.4	gutter-R
70	632	674	3.4	gutter-L
71	697	674	4.5	floor bracket
72	134	679	4.5	socket plug
73	584	679	4.5	socket plug
74	336	705	6.5	M6 insert screw
75	382	705	6.5	M6 insert screw
76	336	718	6.5	M6 insert screw
77	382	718	6.5	M6 insert screw

#	x	y	Hole	For
78	87	734	3.4	gutter-R
79	343	737	3.2	motor-relief
80	632	734	3.4	gutter-L
81	64	772	3.4	corner-R
82	375	770	3.2	motor-relief
83	654	772	3.4	corner-L
84	209	774	5.5	M5 insert screw
85	509	774	5.5	M5 insert screw
86	209	822	4.5	floor bracket
87	509	822	4.5	floor bracket

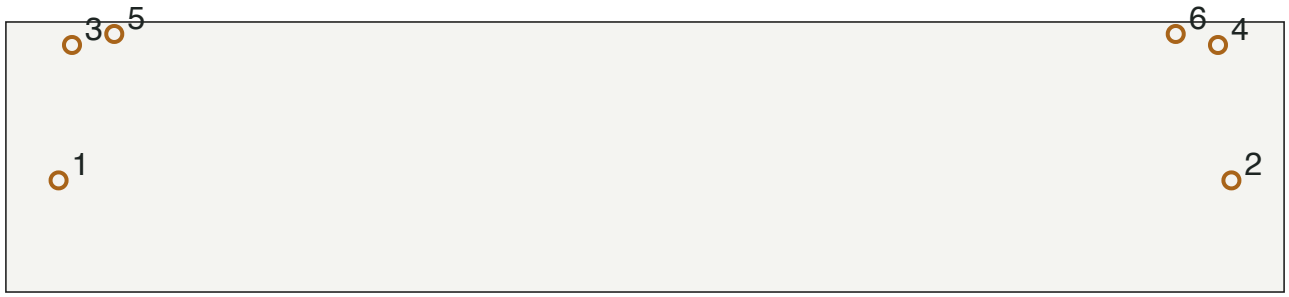
Base front wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	39	55	4.5	corner-L
2	599	55	4.5	corner-R
3	33	124	4.5	corner bracket
4	605	124	4.5	corner bracket
5	54	129	4.5	leg holder
6	584	129	4.5	leg holder

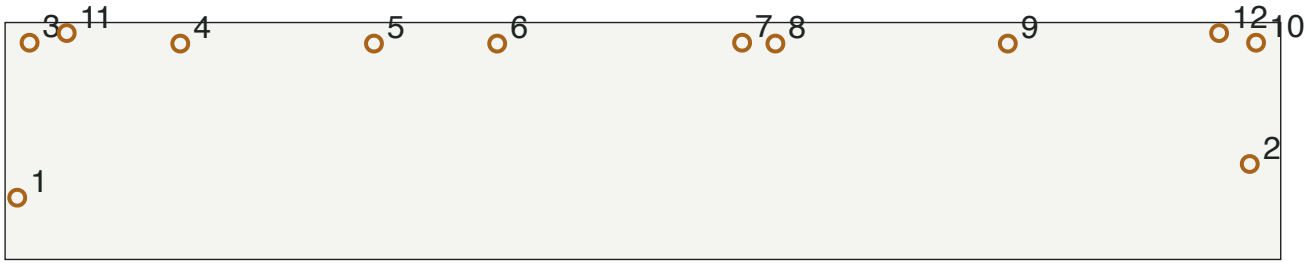
Base rear wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	26	56	4.5	corner-R
2	612	56	4.5	corner-L
3	33	124	4.5	corner bracket
4	605	124	4.5	corner bracket
5	54	129	4.5	leg holder
6	584	129	4.5	leg holder

Base side wall, left

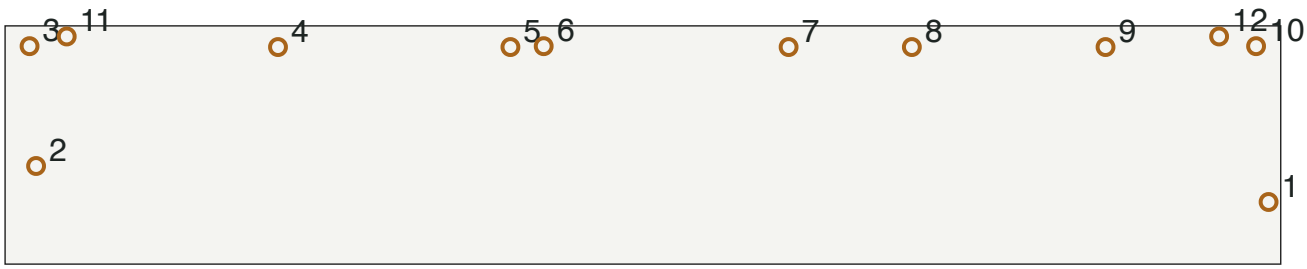


0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	7	35	4.5	corner-L
2	707	54	4.5	corner-L
3	14	124	4.5	corner bracket
4	100	123	4.5	saddle
5	210	123	4.5	saddle
6	280	123	4.5	saddle
7	419	124	4.5	corner bracket
8	438	123	4.5	saddle
9	570	123	4.5	saddle

#	x	y	Hole	For
10	711	124	4.5	corner bracket
11	35	129	4.5	leg holder
12	690	129	4.5	leg holder

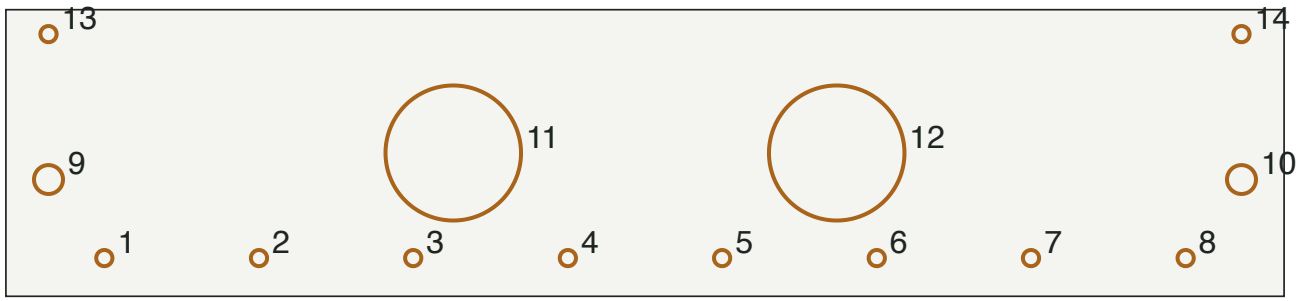
Base side wall, right



0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	718	35	4.5	corner-R
2	18	56	4.5	corner-R
3	14	124	4.5	corner bracket
4	155	123	4.5	saddle
5	287	123	4.5	saddle
6	306	124	4.5	corner bracket
7	445	123	4.5	saddle
8	515	123	4.5	saddle
9	625	123	4.5	saddle
10	711	124	4.5	corner bracket
11	35	129	4.5	leg holder
12	690	129	4.5	leg holder

Base partition



0,0 · 600 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	46	18	3.2	split-gasket-groove
2	119	18	3.2	split-gasket-groove
3	191	18	3.2	split-gasket-groove
4	264	18	3.2	split-gasket-groove
5	336	18	3.2	split-gasket-groove
6	409	18	3.2	split-gasket-groove
7	481	18	3.2	split-gasket-groove
8	554	18	3.2	split-gasket-groove
9	20	55	13.5	1/2-13 bolt into coupler
10	580	55	13.5	1/2-13 bolt into coupler
11	210	68	63.5	hole saw
12	390	68	63.5	hole saw
13	20	124	4.5	corner bracket
14	580	124	4.5	corner bracket

Base bridge y=-190



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	599	12	4.5	core post
3	520	12	4.5	core post

#	x	y	Hole	For
4	325	12	4.5	core post
5	130	12	4.5	core post
6	52	12	4.5	core post
7	16	11	4.5	saddle

Base bridge y=-58



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	489	12	4.5	core post
3	161	12	4.5	core post
4	16	11	4.5	saddle

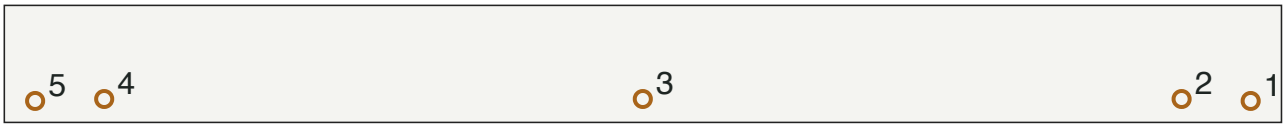
Base bridge y=100



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	565	12	4.5	core post
3	489	12	4.5	core post
4	161	12	4.5	core post
5	85	12	4.5	core post
6	16	11	4.5	saddle

Base bridge y=170



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	600	12	4.5	core post
3	325	12	4.5	core post
4	51	12	4.5	core post
5	16	11	4.5	saddle

Base bridge y=280



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	600	12	4.5	core post
3	480	12	4.5	core post
4	170	12	4.5	core post
5	51	12	4.5	core post
6	16	11	4.5	saddle

Base leg bridge y=-310



0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	426	12	4.5	core post
3	54	12	4.5	core post
4	6	11	4.5	leg holder

Base leg bridge y=344

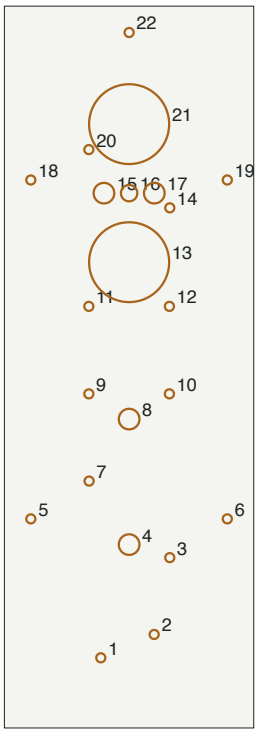


0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	385	12	4.5	core post
3	95	12	4.5	core post
4	6	11	4.5	leg holder

Uprights: hole maps

Upright floor (x2)

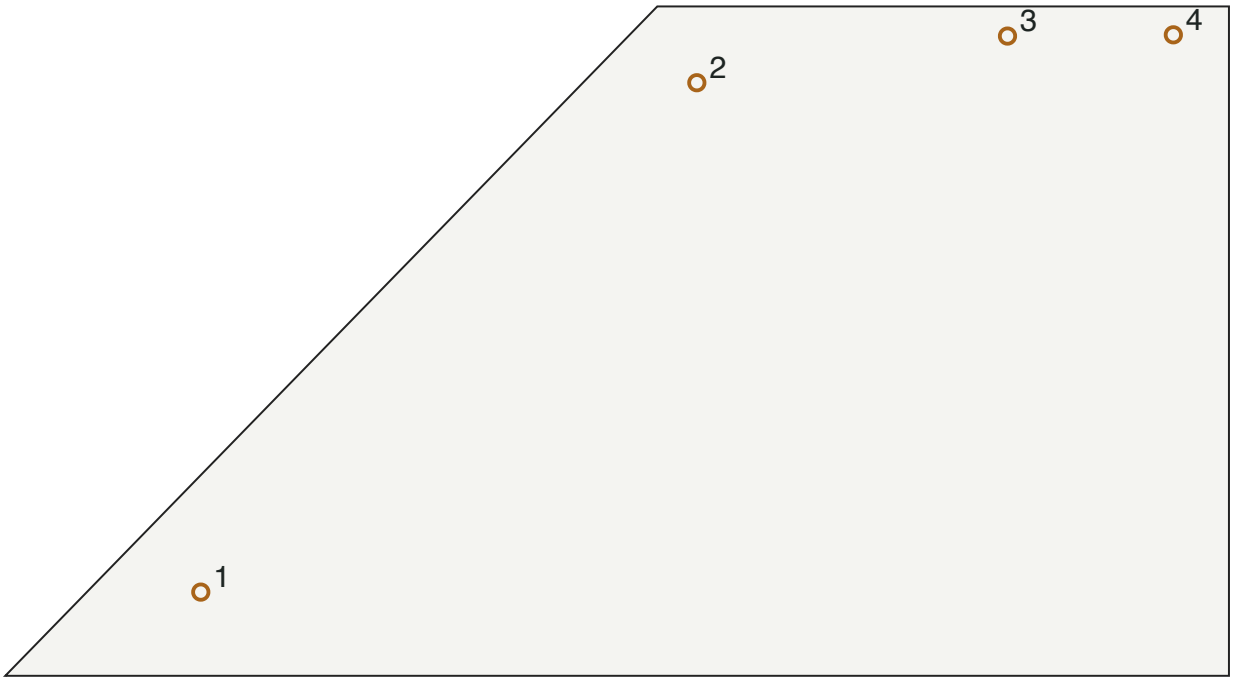


0,0 · 198 x 575 mm · seen from the

#	x	y	Hole	For
1	77	56	4.5	toe-filler
2	119	75	4.5	toe-filler
3	131	136	3.2	foot-gasket-groove
4	99	146	16.3	5/8 core rod
5	21	167	4.5	floor bracket
6	177	167	4.5	floor bracket
7	67	197	3.2	foot-gasket-groove
8	99	246	16.3	5/8 core rod
9	67	266	3.2	foot-gasket-groove
10	131	266	3.2	foot-gasket-groove
11	67	336	3.2	foot-gasket-groove
12	131	336	3.2	foot-gasket-groove
13	99	371	63.5	hole saw
14	131	414	3.2	foot-gasket-groove
15	79	426	16.3	5/8 core rod
16	99	426	13.1	1/2-13 bolt, base dowel hole

#	x	y	Hole	For
17	119	426	16.3	5/8 core rod
18	21	437	4.5	floor bracket
19	177	437	4.5	floor bracket
20	67	461	3.2	foot-gasket-groove
21	99	481	63.5	hole saw
22	99	554	4.5	floor bracket

Upright side plate, outer (x2)

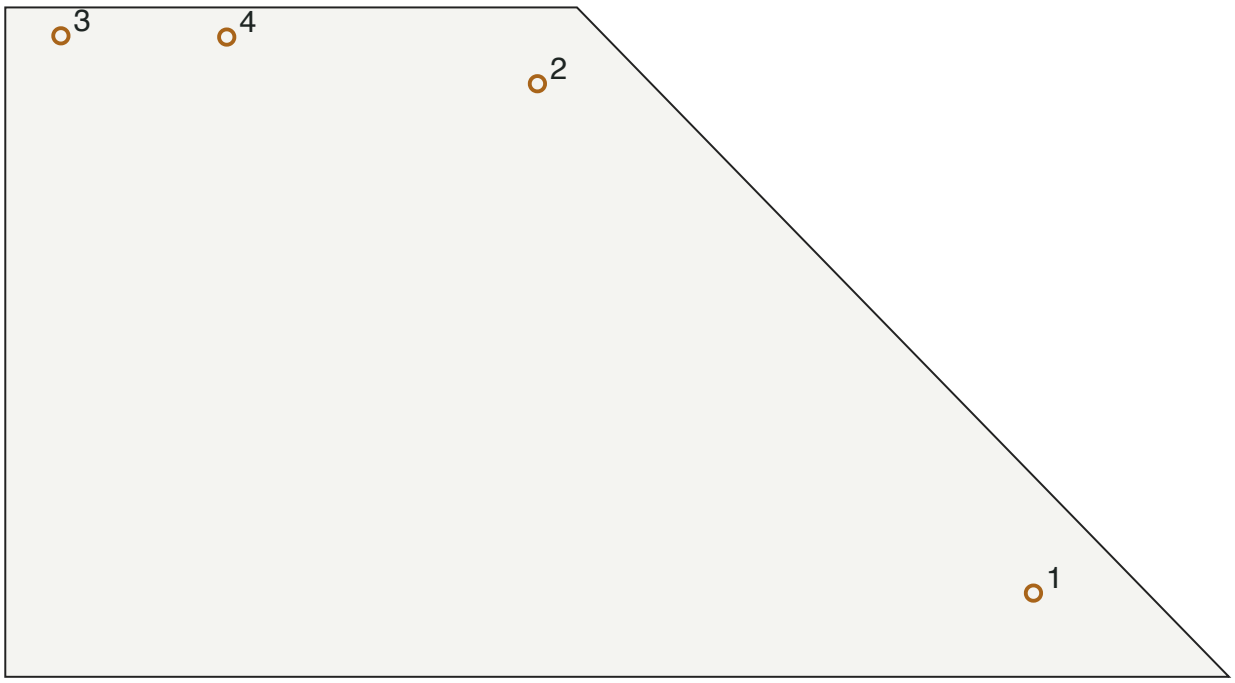


0,0 · 495 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	79	34	4.5	corner bracket
2	280	240	4.5	corner bracket
3	406	259	4.5	saddle
4	473	260	4.5	corner bracket

Outline corners (x, y): (0, 0), (495, 0), (495, 271), (264, 271)

Upright side plate, inner (x2)



0,0 · 495 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	416	34	4.5	corner bracket
2	215	240	4.5	corner bracket
3	22	260	4.5	corner bracket
4	90	259	4.5	saddle

Outline corners (x, y): (495, 0), (0, 0), (0, 271), (231, 271)

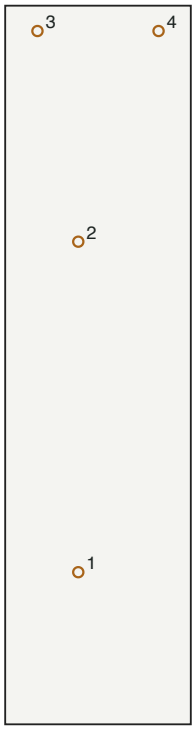
Upright back board (x2)



0,0 · 80 x 271 mm · seen from t

#	x	y	Hole	For
1	3	260	4.5	corner bracket
2	77	260	4.5	corner bracket

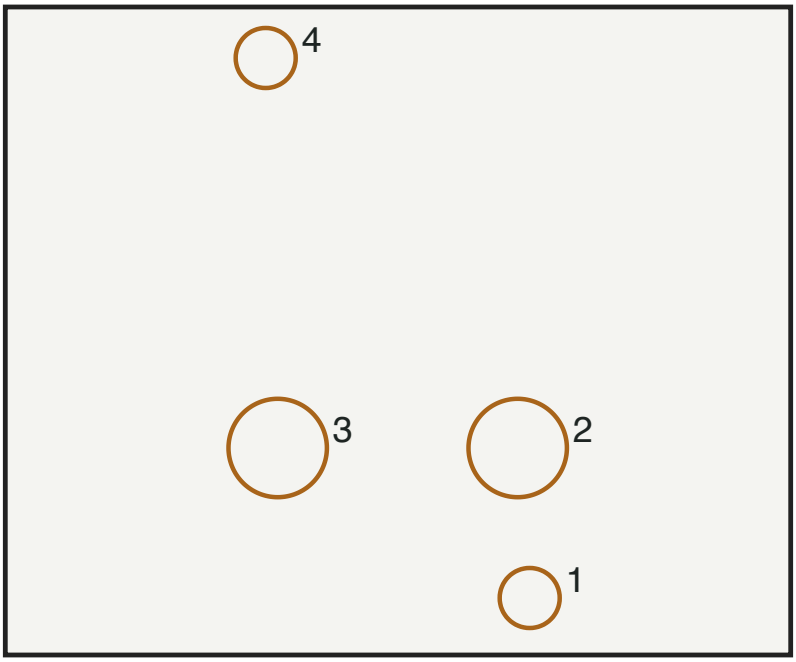
Upright front board (x2)



0,0 · 80 x 312 mm · seen frc

#	x	y	Hole	For
1	32	66	4.5	slope-pocket--60
2	32	209	4.5	slope-pocket-40
3	14	301	4.5	corner bracket
4	66	301	4.5	corner bracket

Upright bridge (x2)

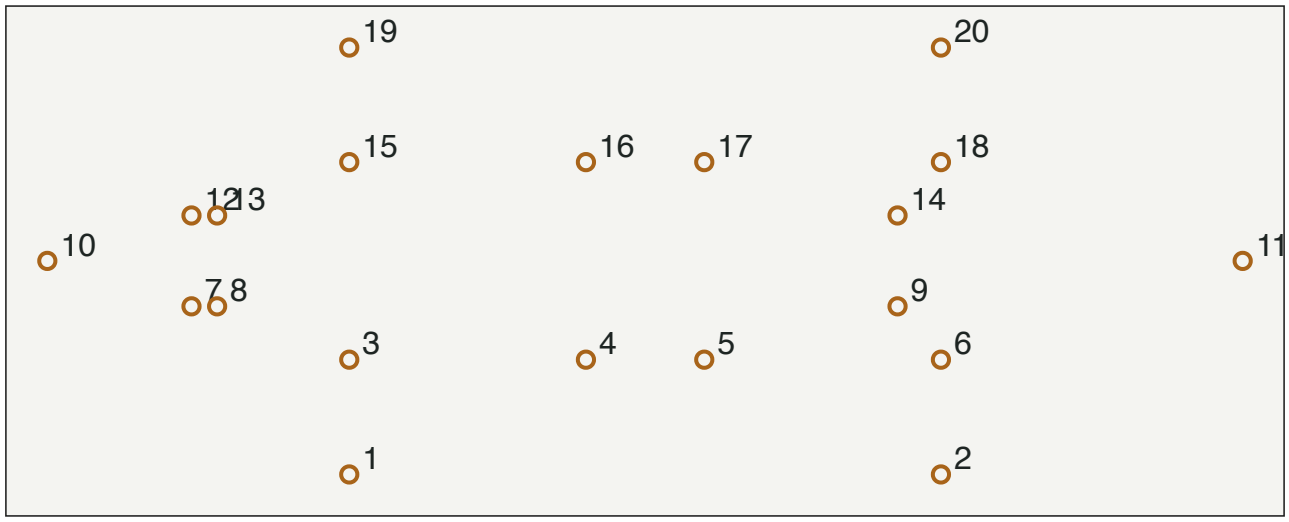


0,0 · 131 x 108 mm · seen from the outside (top)

#	x	y	Hole	For
1	87	10	10	3/8-16 bolt into the top plate
2	85	34	16.3	5/8 core rod
3	45	34	16.3	5/8 core rod
4	43	100	10	3/8-16 bolt into the top plate

Beam: hole maps

Beam floor



0,0 · 648 x 258 mm · seen from the cast side

#	x	y	Hole	For
1	174	21	4.5	floor bracket
2	474	21	4.5	floor bracket
3	174	79	5.5	M5 insert screw
4	294	79	5.5	M5 insert screw
5	354	79	5.5	M5 insert screw
6	474	79	5.5	M5 insert screw
7	94	106	6.5	M6 insert screw
8	107	106	6.5	M6 insert screw
9	452	106	6.5	M6 insert screw
10	21	129	4.5	floor bracket
11	627	129	4.5	floor bracket
12	94	152	6.5	M6 insert screw
13	107	152	6.5	M6 insert screw
14	452	152	6.5	M6 insert screw
15	174	179	5.5	M5 insert screw
16	294	179	5.5	M5 insert screw
17	354	179	5.5	M5 insert screw
18	474	179	5.5	M5 insert screw
19	174	237	4.5	floor bracket

#	x	y	Hole	For
20	474	237	4.5	floor bracket

Beam top wall



0,0 · 568 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	487	25	13.1	1/2 core rod
4	531	25	13.1	1/2 core rod
5	37	115	13.1	1/2 core rod
6	81	115	13.1	1/2 core rod
7	487	115	13.1	1/2 core rod
8	531	115	13.1	1/2 core rod
9	84	133	4.5	back-edge-1
10	219	133	4.5	back-edge-1
11	351	133	4.5	back-edge-1
12	484	133	4.5	back-edge-1
13	33	154	4.5	corner bracket
14	535	154	4.5	corner bracket

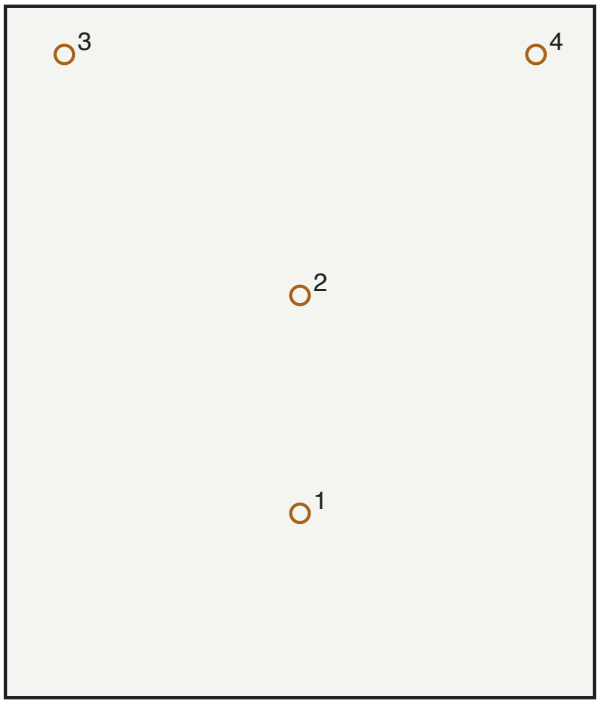
Beam bottom wall



0,0 · 568 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	487	25	13.1	1/2 core rod
4	531	25	13.1	1/2 core rod
5	59	70	9.6	3/8 dowel
6	509	70	9.6	3/8 dowel
7	37	115	13.1	1/2 core rod
8	81	115	13.1	1/2 core rod
9	487	115	13.1	1/2 core rod
10	531	115	13.1	1/2 core rod
11	33	154	4.5	corner bracket
12	535	154	4.5	corner bracket

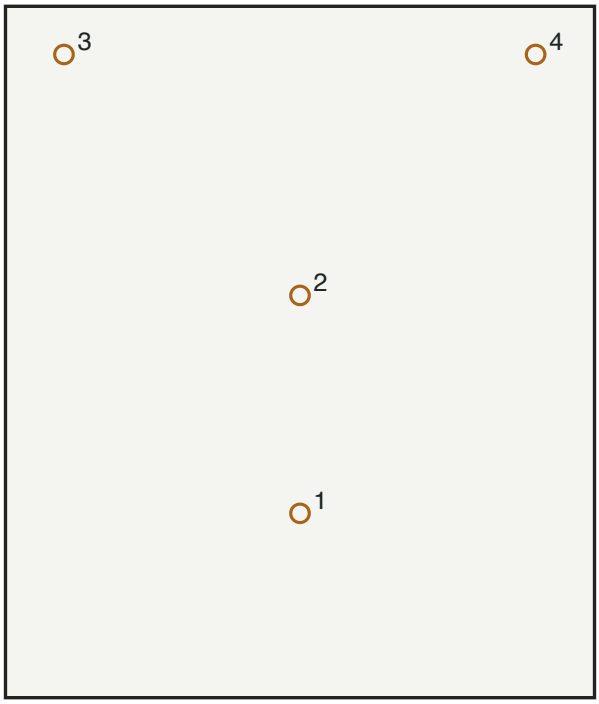
Beam end wall, left



0,0 · 140 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	70	44	4.5	tube locator
2	70	96	4.5	tube locator
3	14	154	4.5	corner bracket
4	126	154	4.5	corner bracket

Beam end wall, right



0,0 · 140 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	70	44	4.5	tube locator
2	70	96	4.5	tube locator
3	14	154	4.5	corner bracket
4	126	154	4.5	corner bracket

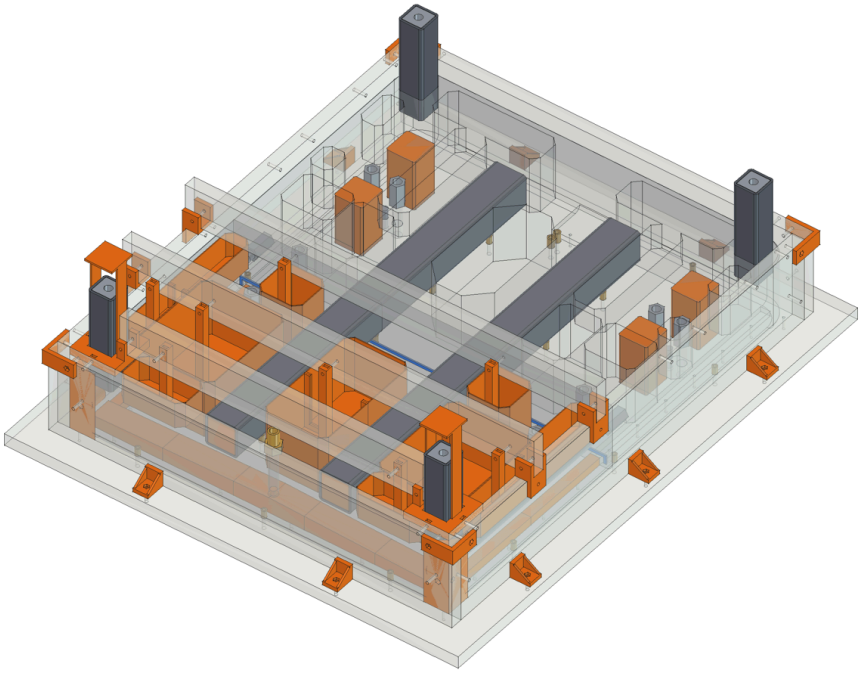
Putting the moulds together

Base: both halves on one floor

The rear half is cast first, against a partition at the split. Then the partition comes out and the front half is cast against the rear half's waxed split face, on the same floor. The two top faces come out flush, the split faces match, and the front tubes slide straight onto the rear half's spigots.

1. Drill the floor, walls, partition and bridges. Wax and PVA every face that the mix touches.
2. Stand the floor on two or three battens (the insert screws and floor bolts come up from underneath).
3. Stand the walls on the floor. Fit the four corner brackets at the top corners, then the floor brackets along every wall foot. Square the box with a tape across the diagonals before tightening.
4. Fit the partition between the side walls with its two brackets on the front side, and the two struts from the front wall to its foot. Push the gasket-groove beads onto its pins; push the spigot bushings into its hole-saw holes from the front.

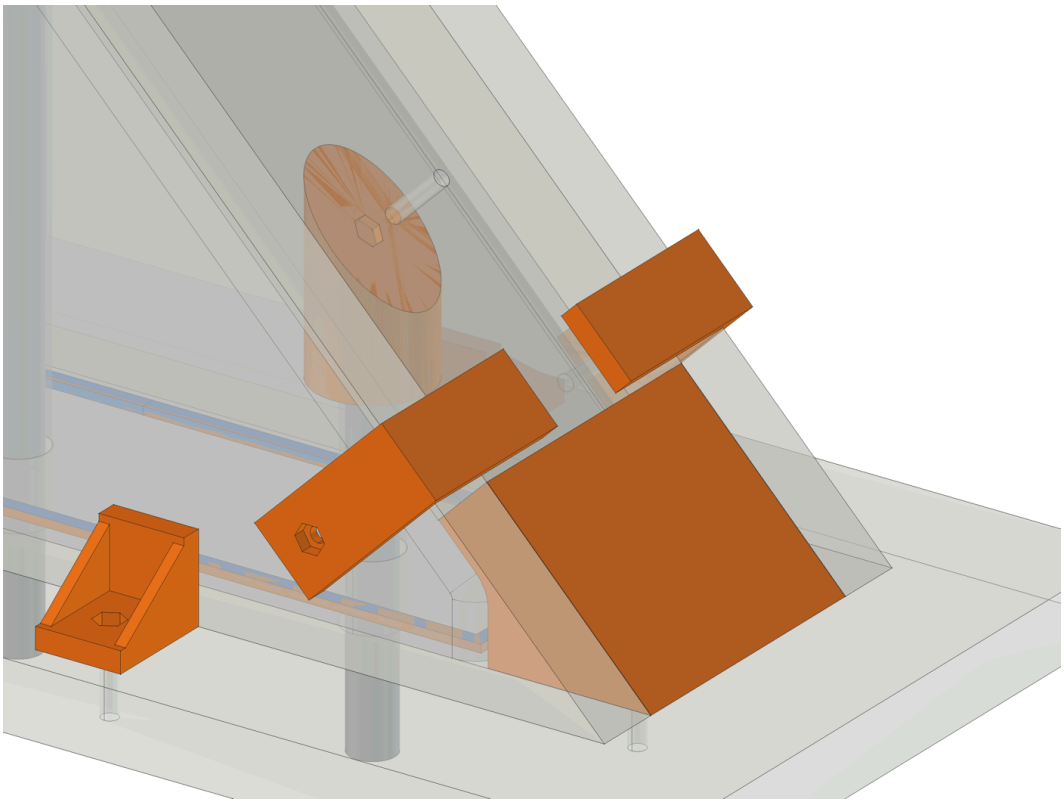
5. Bolt the rear-half pieces on: the gutter formers and the motor relief to the floor, the corner fillers to the floor and walls, the socket plugs to the floor (each with its liner over it), the inserts and the tie-rod couplers on their greased screws and bolts, the base dowels in their holes.
6. Hold the draw-rod couplers on greased 1/2-13 bolts through the partition. Slide the spigots through the bushings and put the rear tubes on them, with their sleeves and end plugs.
7. Bolt the leg holders into the rear corners, drop the legs through them and put a sleeve on each leg.
8. Cast the 20 mm skin, set the cores on it (sand in them), drop the bridges into the saddles and bolt each core's post to its bridge. Fill to the line.
9. After the cure: unbolt the bridges, partition and struts, leave the rest. Pull the gasket-groove beads, cover the groove with packing tape, and wax and PVA the rear half's split face and the spigots.
10. Front half the same way: formers, trough, corner fillers, couplers and inserts; front tubes onto the spigots; draw rods screwed into the rear half's couplers with their ends in the corner fillers; legs; skin, cores, bridges, fill.
11. After the cure, unbolt everything and split the halves (a few wedges at the split, gently).



Uprights (two moulds, cast together)

1. Drill the boards; wax and PVA. Push the post bushings into the floor from below.
2. Stand each side plate on the floor with its floor brackets, the back board between them with its floor bracket, and the two back brackets at the top.
3. Bolt the toe filler to the floor between the side plates, set the front board on it and fit the front brackets: two at its top, two at its foot (those only bolt to the side plates).
4. Pin the gasket-groove bead to the floor. Set the mould on its four printed feet.
5. Posts up through the bushings onto the bench; greased 5/8 in rods up through the floor: the two short ones into the slope plugs (bolted to the front board), the two long ones on through the top. The dowel-hole bolt with its nut from underneath.

6. Fill to the line and vibrate. Then slide the top pocket plugs down the long rods, set the top plate (greased 3/8 bolts through the bridge and the spacers into it), drop the bridge into its saddles.



Beam

1. Drill the boards; wax and PVA. Walls on the floor with the floor brackets, corner brackets at the top.
2. Screw the support pads and riser inserts to the floor on greased screws. Push the dowels into the bottom wall onto their printed stops.
3. Bolt the tube locators through the end walls (M4 x 70 into the nut in each square end) and hang the tubes between them. Bolt the back-edge fillers to the top wall.
4. Greased 1/2 in rods through the top and bottom walls for the eight bolt holes. Fill to the line.
5. The locators leave a 16 mm hole in each beam end; plug it with a dark printed cap or leave it.